Windows Threat Detection 1 — Incident Investigation Report

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Investigation Type : Windows Threat Detection / SOC Investigation / Threat Hunting

Platform: Windows

Lab : TryHackMe — Windows Threat Detection 1

Date: September 2026

Executive Summary

This report documents the investigation performed during the Windows Threat Detection 1 laboratory. The objective of the investigation was to analyze Windows Security logs, Sysmon telemetry, and endpoint activity to identify common Initial Access techniques used against Windows systems.

The investigation focused on:

1. Remote Desktop Protocol (RDP) attacks
2. Phishing-based initial access
3. Malicious file execution
4. Suspicious network communication
5. USB-based malware propagation
6. Process execution telemetry

Analysis of Windows logs and Sysmon events enabled reconstruction of attacker behavior and demonstrated how SOC analysts detect early-stage compromise activity using native Windows logging.

1. Investigation Objectives

The primary objectives of the investigation were:

- Detect Initial Access techniques
- Investigate malicious RDP activity

- Analyze phishing-based compromise events
- Investigate suspicious process execution
- Analyze malicious network communication
- Detect USB malware propagation activity
- Reconstruct attacker timeline
- Identify indicators of compromise (IOCs)

2. Log Sources Investigated

2.1 Windows Security Logs

Authentication and account activity were analyzed using:

Event Viewer → Windows Logs → Security

Purpose

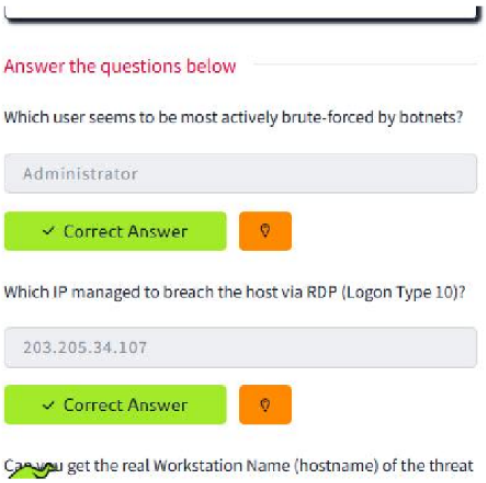
- Failed authentication monitoring
- Successful RDP login investigation
- Logon activity correlation
- Source IP identification

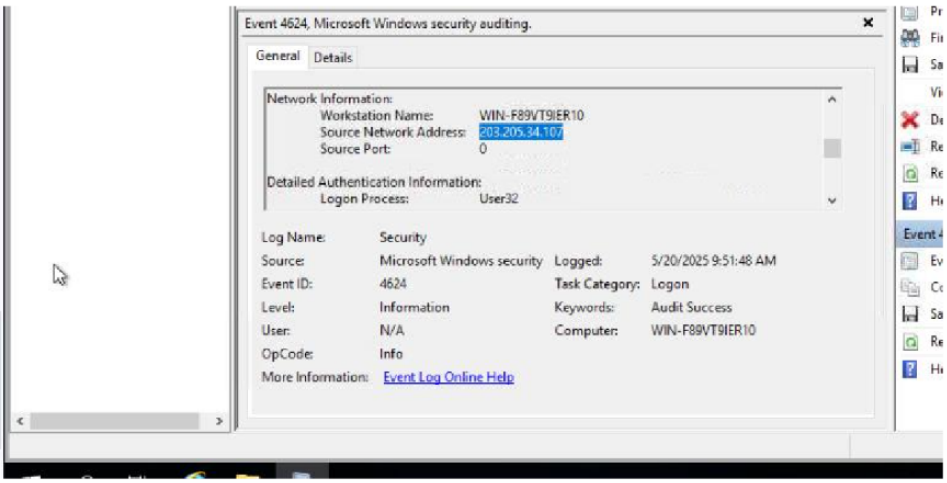
Key Event IDs

4624 — Successful Logon

4625 — Failed Logon

Screenshot Placement





2.2 Sysmon Operational Logs

Endpoint telemetry was analyzed using:

Applications and Services Logs → Microsoft → Windows → Sysmon → Operational

Purpose

- Process creation monitoring
- Network connection analysis
- File creation investigation
- Malware execution detection

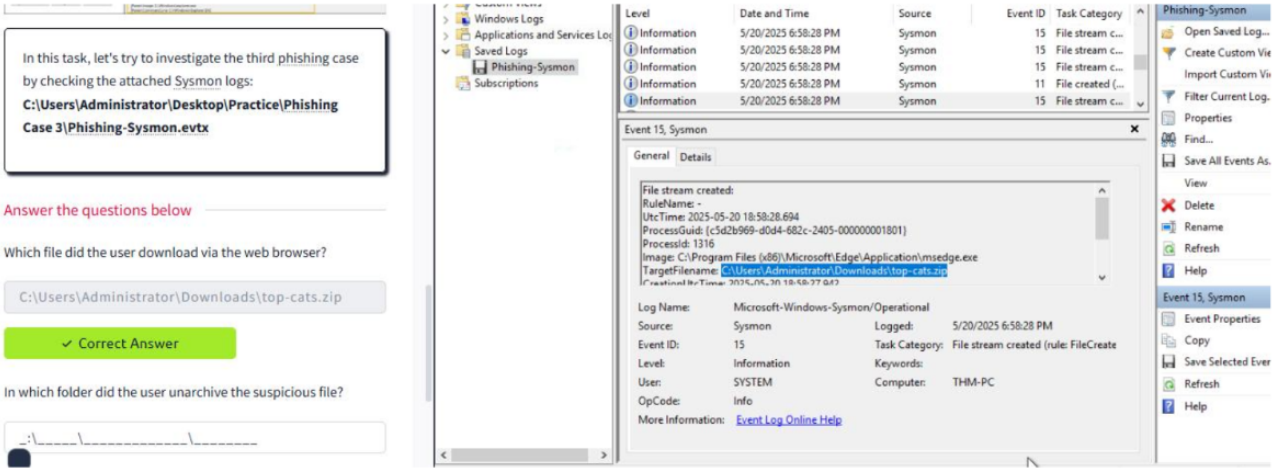
Key Event IDs

Event ID 1 — Process Creation

Event ID 3 — Network Connection

Event ID 11 — File Creation

Screenshot Placement



3. Investigation Methodology

The investigation followed a structured SOC workflow:

1. Review Security logs
2. Identify failed authentication attempts
3. Investigate successful RDP logons
4. Correlate source IP addresses
5. Analyze phishing-related execution events

6. Review Sysmon process creation logs

7. Investigate malicious network communication

8. Analyze USB execution activity

9. Reconstruct activity timeline

10. Document findings and evidence

4. Initial Access Investigation — RDP Activity

4.1 Failed Authentication Attempts

The following Event ID was investigated:

Event ID 4625

Findings

Multiple failed RDP authentication attempts were identified targeting the Administrator account.

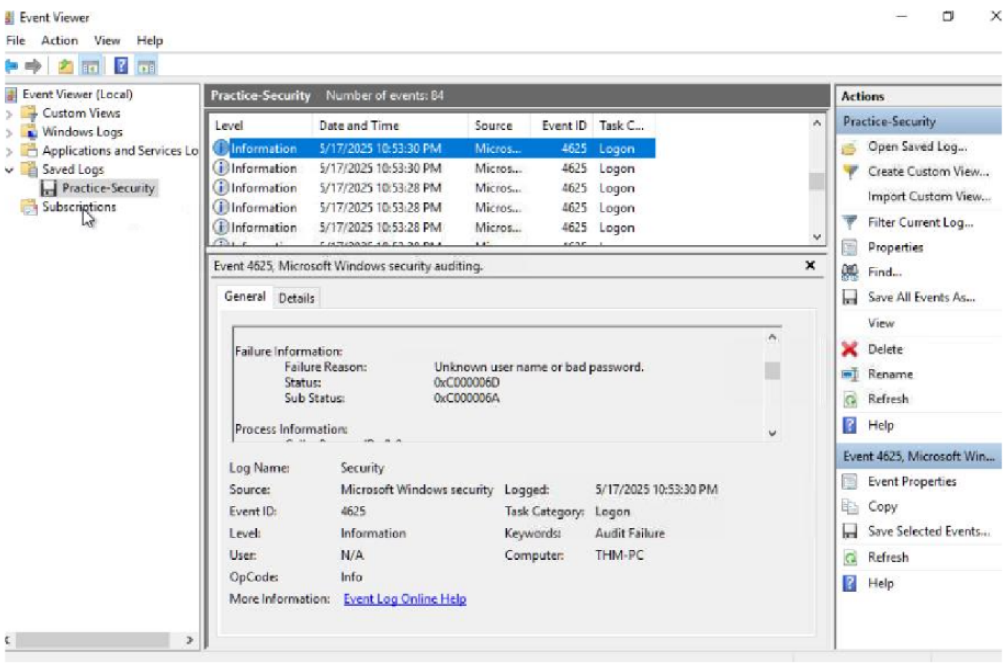
Analysis

- Observed activity is consistent with:
- Brute-force attacks
- Automated credential attacks
- Password spraying activity

Security Impact

Repeated failed authentication attempts indicate active adversary probing against exposed RDP services.

Screenshot Placement



4.2 Successful RDP Authentication

The following Event ID was reviewed:

Event ID 4624

Findings

A successful RDP login was identified from the following IP address:

203.205.34.107

The associated workstation name was identified as:

DESKTOP-QNBC4UU

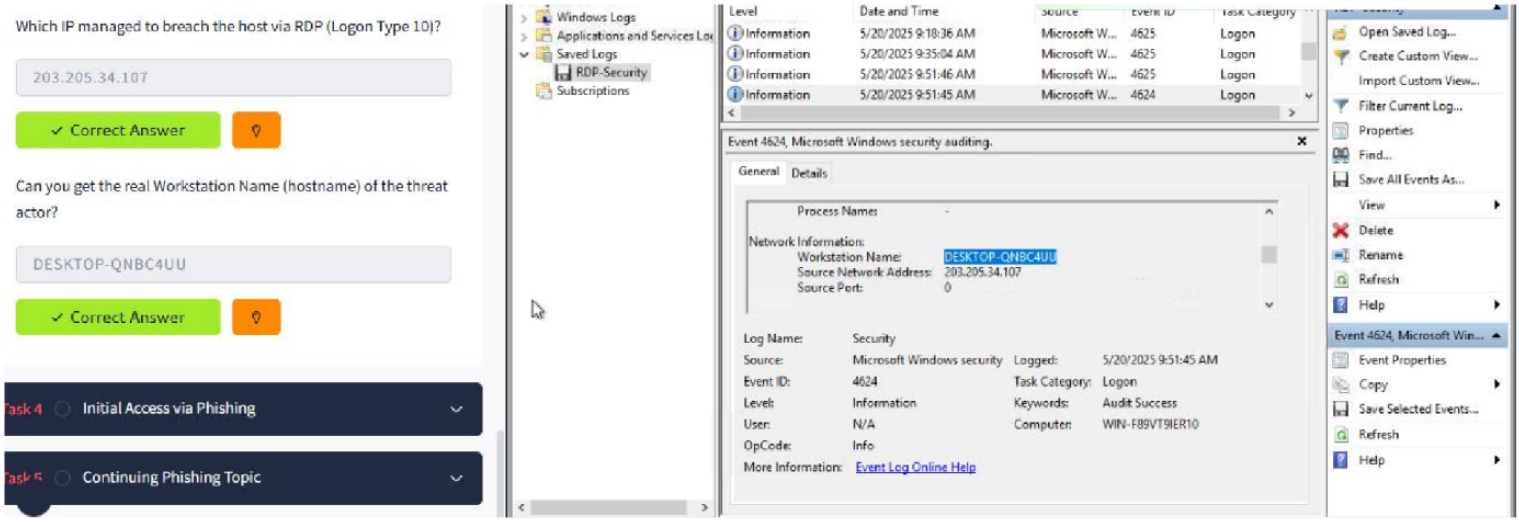
Analysis

The successful authentication following repeated failed attempts suggests unauthorized remote access activity.

Security Impact

Successful remote authentication may provide attackers with an interactive foothold on the compromised system.

Screenshot Placement



5. Initial Access Investigation — Phishing Activity

5.1 Malicious LNK File Execution

Findings

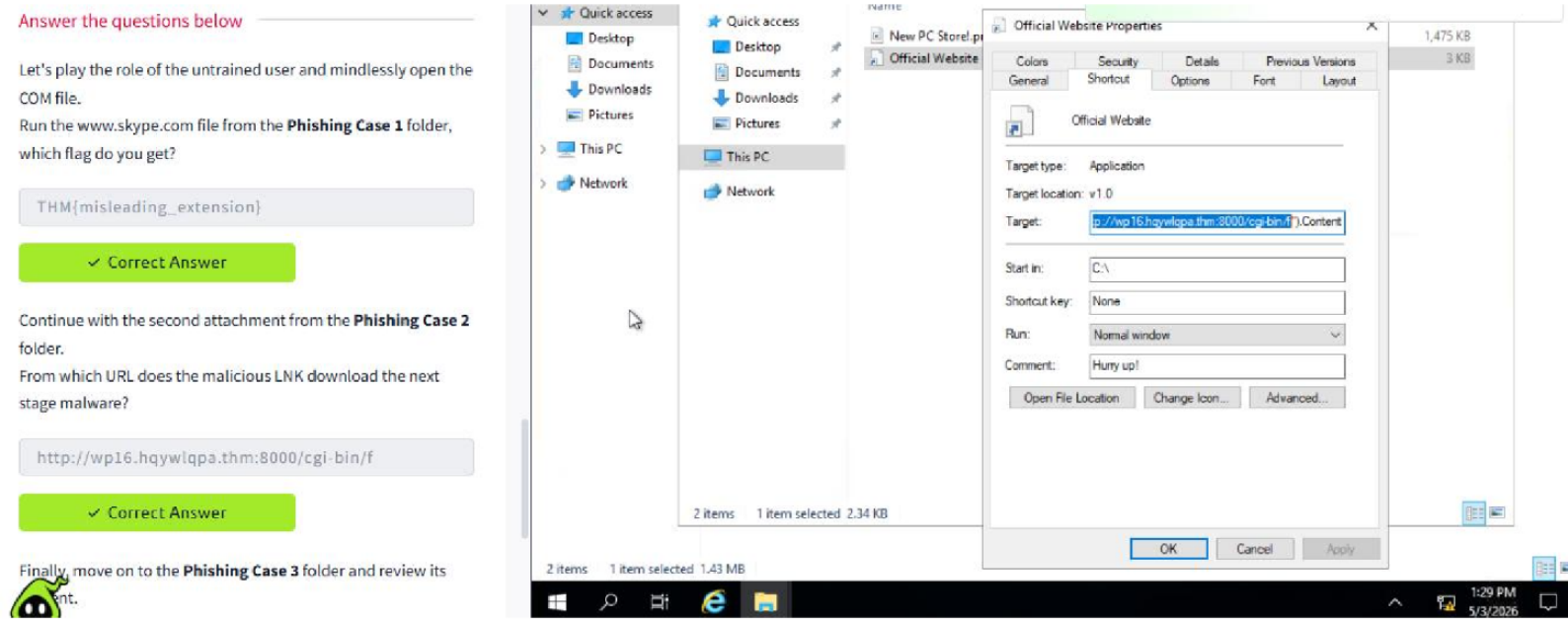
The phishing investigation identified execution of a malicious LNK shortcut file designed to retrieve additional malware payloads.

Analysis

Observed behavior indicates:

- User-assisted malware execution
- Initial payload staging
- Phishing-based compromise activity
- Security Impact
- Malicious shortcut files are commonly used to bypass user suspicion and launch hidden PowerShell or command execution activity.

Screenshot Placement



5.2 Malware Download Investigation

Findings

The malicious shortcut downloaded additional payloads from an external URL.

Analysis

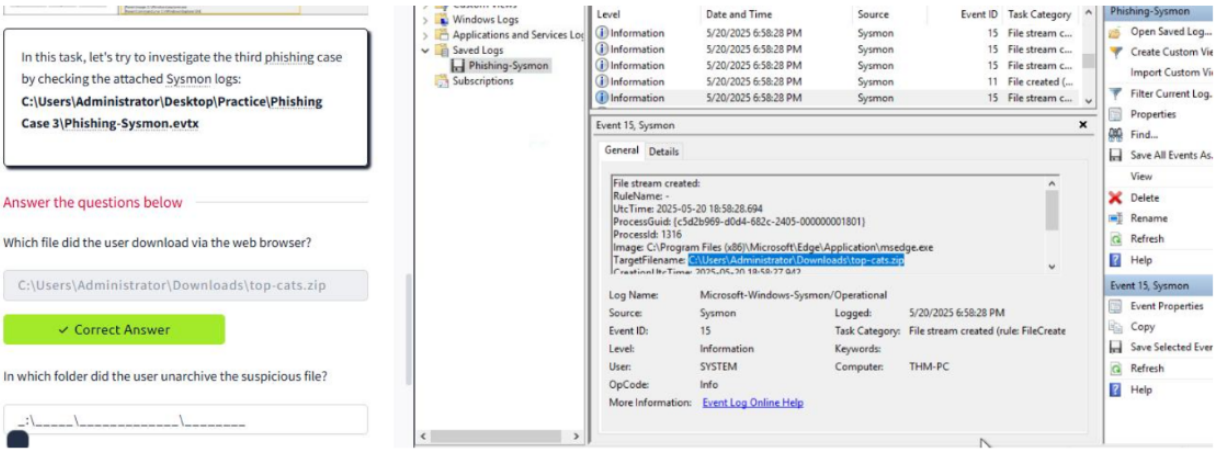
Observed behavior suggests:

- Malware staging activity
- External payload retrieval
- User compromise through phishing

Security Impact

Downloaded payloads may introduce malware capable of persistence or credential theft.

Screenshot Placement



5.3 Suspicious File Extraction Activity

Findings

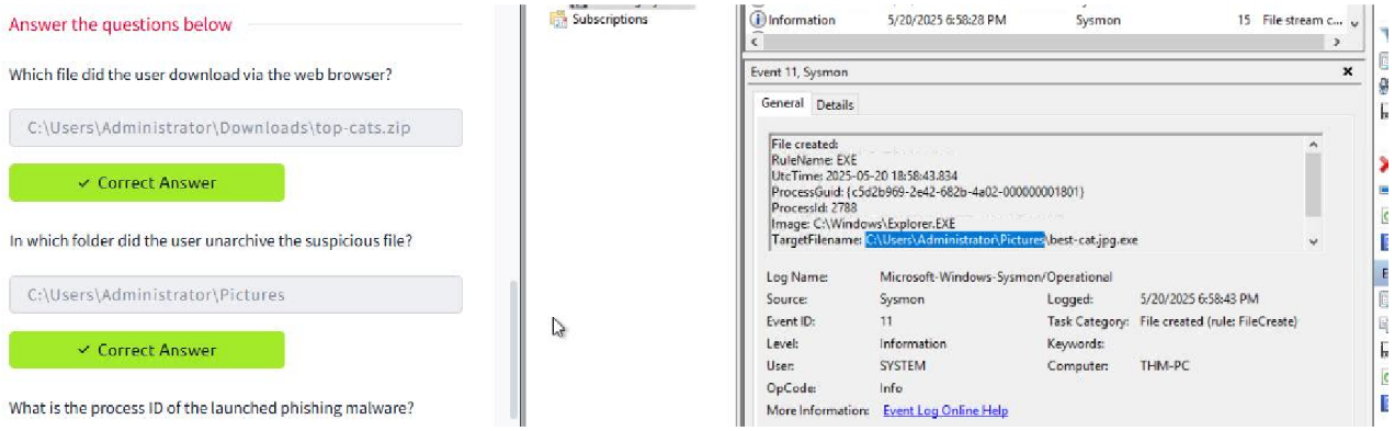
The investigation identified extraction of suspicious archived content into a user-accessible directory.

Analysis

Archive extraction activity is frequently associated with:

- Malware unpacking
- Payload staging
- User-driven compromise workflows

Screenshot Placement



6. Sysmon Investigation

6.1 Process Creation Analysis

The following Sysmon event was analyzed:

Sysmon Event ID 1

Findings

Process creation telemetry revealed execution of suspicious binaries and phishing-related processes.

Analysis

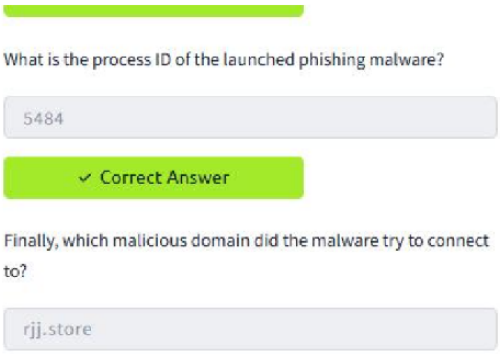
The telemetry provided:

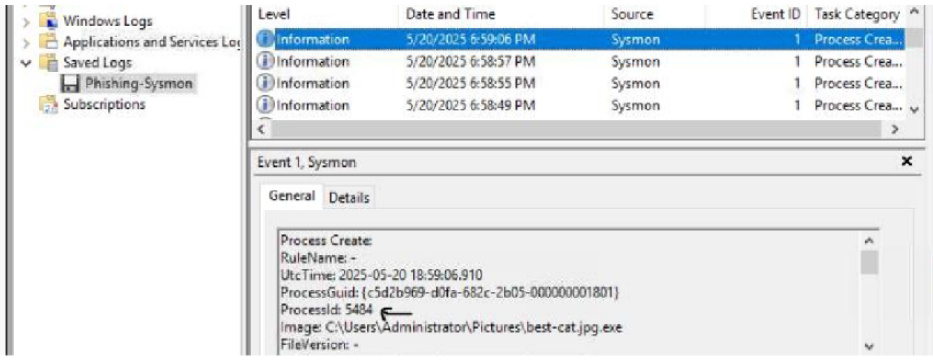
- Process names
- Parent-child relationships
- Command-line execution details
- User context information

Security Impact

Process creation monitoring enables SOC analysts to reconstruct attacker execution chains and detect malicious activity.

Screenshot Placement





6.2 Network Connection Analysis

The following Sysmon event was reviewed:

Sysmon Event ID 3

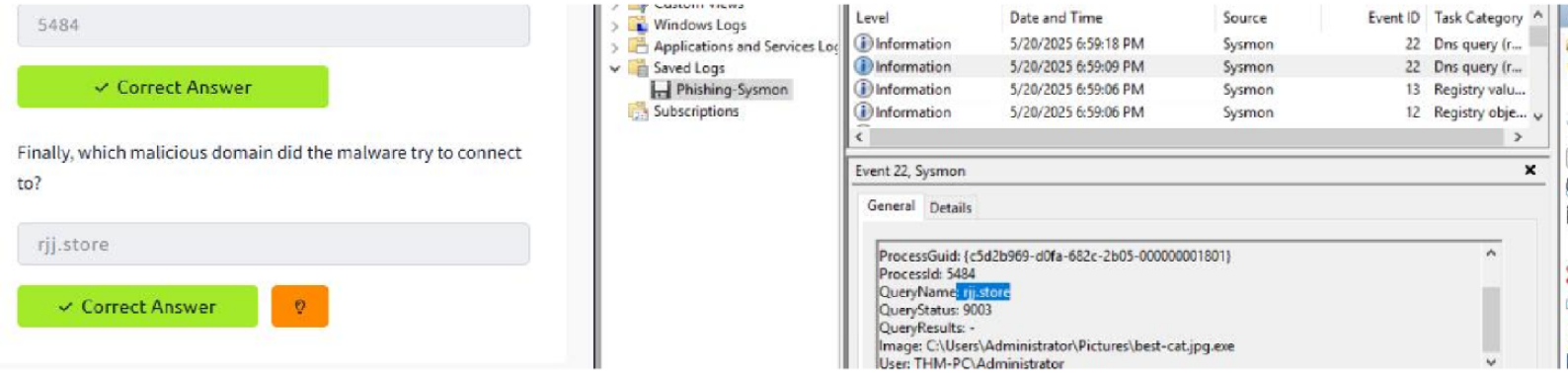
Findings

Outbound network communication to suspicious infrastructure was identified during malware execution.

Analysis

- Observed network activity indicates:
- Malware communication attempts
- External infrastructure interaction
- Potential payload retrieval activity

Screenshot Placement



7. USB-Based Initial Access Investigation

7.1 USB Malware Execution

Findings

The investigation identified suspicious executable activity launched from a connected USB device.

Analysis

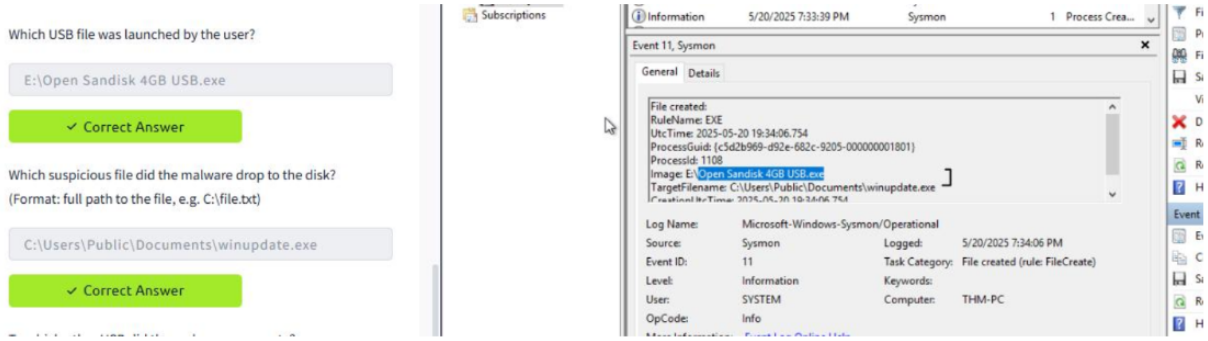
Observed behavior indicates:

- Removable media malware execution
- User-assisted compromise activity
- Potential malware propagation behavior

Security Impact

USB-delivered malware remains an effective Initial Access vector in enterprise environments.

Screenshot Placement



7.2 Malware Propagation Activity

Findings

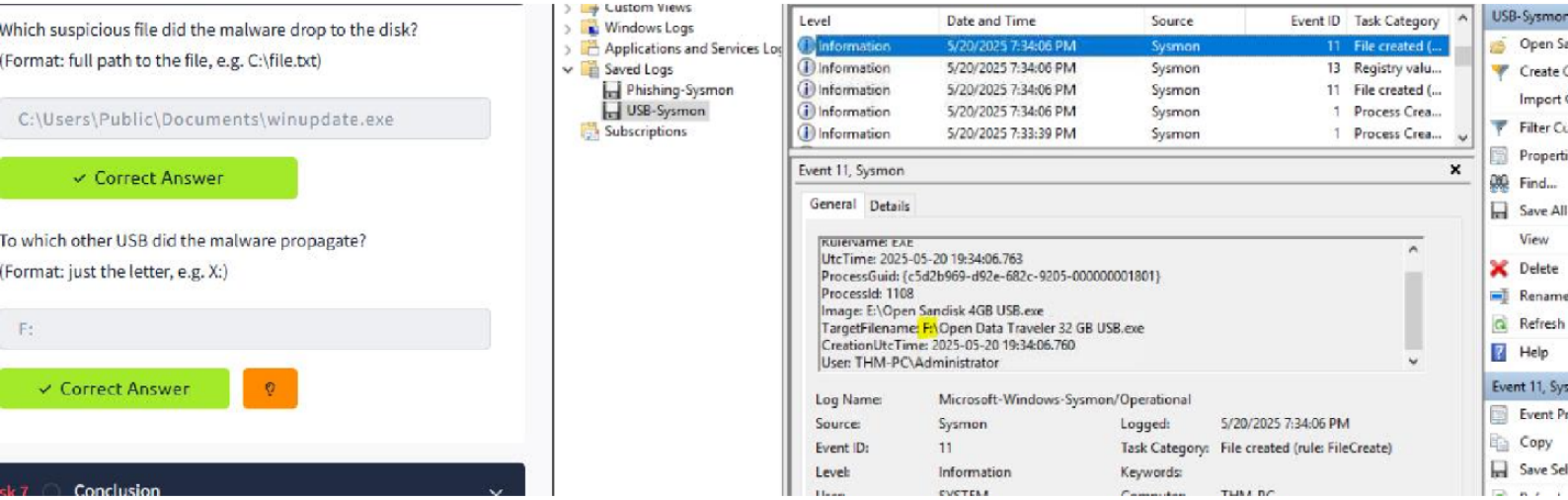
The malware attempted propagation to additional removable media devices.

Analysis

Propagation behavior may indicate:

- Worm-like functionality
- Automated malware spreading
- Removable media abuse

Screenshot Placement



8. Indicators of Compromise (IOCs)

Network Indicators

203.205.34.107 ⟶ Suspicious RDP source IP

Malicious external domains Payload retrieval infrastructure

Process Indicators

Malicious LNK file ⟶ Phishing payload launcher

Suspicious executable ⟶ Malware execution artifact

powershell.exe ⟶ Command execution utility

9. Behavioral Indicators

- Repeated failed RDP authentication attempts
- Successful RDP login after brute-force activity
- Malicious shortcut execution
- Suspicious payload download activity
- Unusual process execution
- Suspicious outbound network communication
- USB-based malware activity

10. Recommendations

- Immediate Actions
- Disable exposed RDP access
- Reset compromised credentials
- Remove malicious files
- Block malicious IP addresses
- Isolate affected systems
- Scan removable media devices

11. Long-Term Improvements

- Implement MFA for RDP access
- Deploy Sysmon enterprise-wide
- Centralize endpoint logs in SIEM
- Harden PowerShell execution policies
- Restrict USB device usage
- Improve phishing awareness training
- Monitor suspicious Event IDs continuously

12. Detection Opportunities

RDP Brute Force Detection

Monitor for:

High volumes of Event ID 4625 failures

Successful 4624 logons after failures

Phishing Detection

Monitor for:

Suspicious LNK execution

PowerShell launched from Office or shortcut files

USB Monitoring

Monitor for:

Executables launched from removable media

Unexpected file replication activity

13. Conclusion

The investigation demonstrated how Windows Security logs and Sysmon telemetry can be used to detect and investigate common Windows Initial Access techniques.

Analysis of authentication events, phishing execution activity, suspicious process creation, outbound communication, and USB-based malware

activity enabled reconstruction of attacker behavior throughout the investigation timeline.

The laboratory successfully demonstrated:

- RDP attack investigation
- Phishing detection workflows
- Sysmon-based endpoint analysis
- Malware execution monitoring
- USB threat detection
- SOC investigation methodology

14. Analyst Notes

1. This investigation demonstrated practical experience in:
2. Windows Event Log analysis
3. Sysmon telemetry analysis
4. Initial Access detection
5. Phishing investigation
6. RDP threat hunting
7. USB malware investigation
8. Timeline reconstruction
9. Incident reporting